REPLACEMENT ABSTRACT

Application No.: 19/639,222

Art Unit: Pre-Exam / TC 2100

Examiner: Andrea Bennett

A MEMORY SYSTEM FOR THE STORAGE AND RETRIEVAL OF PARAMETRIC AND SYMBOLIC DATA REPRESENTATIONS

ABSTRACT

A memory system and architecture designed for the storage and retrieval of data as parametric or symbolic representations rather than as fixed-width, evaluated numerical values. The system comprises a memory controller and a storage medium (such as DRAM, flash, or ROM) where data is stored as variable-length structures representing mathematical 'recipes' or expressions. The memory controller manages the dynamic allocation of physical memory to accommodate these structures and interfaces with a parametric data bus. This architecture ensures that the precision of data is not lost when it is moved from a parametric CPU to memory, enabling the persistence of perfectly represented fractional, irrational, and large-magnitude numbers throughout a complete computing system.

REPLACEMENT CLAIMS

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What is claimed is:

1. A memory system for storing data, comprising:

a physical storage medium; and

a parametric memory controller configured to:

- receive a data structure representing a mathematical expression of a number;
- dynamically allocate a variable-length block of the physical storage medium to store said data structure; and
- retrieve the complete, unmodified data structure from the storage medium upon a read request.

2. The memory system of claim 1, wherein the data structure is received from a parametric central processing unit over a parametric data bus.

3. The memory system of claim 1, wherein the mathematical expression represents a fractional number, and said representation is stored without loss of precision.

4. The memory system of claim 1, wherein the mathematical expression is a symbolic representation of a mathematical or physical constant.

5. The memory system of claim 1, wherein the parametric memory controller manages an addressing index to locate the variable-length blocks within the physical storage medium.

6. The memory system of claim 1, wherein the physical storage medium is volatile random-access memory (RAM).

7. The memory system of claim 1, wherein the physical storage medium is non-volatile memory.

8. The memory system of claim 7, wherein the physical storage medium is read-only memory (ROM) pre-configured with a library of data structures representing mathematical constants.

9. A method for storing data, comprising:

receiving, at a memory controller, a data structure representing a mathematical expression;

determining the size of the received data structure;

allocating a block of memory in a storage medium, wherein the size of the block is based on the determined size of the data structure;

storing the entire data structure in the allocated block; and

creating an index entry that maps a memory address to the location of the allocated block.

Respectfully submitted,

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